What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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Abstract

Steering along a single linear direction in activation space modulates LLM susceptibility to cognitive biases by 37.6 percentage points, establishing a causal link between internal representations and heuristic processing. We identify this direction by comparing matched-architecture model pairs that share weights but differ in reasoning training—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo-3-7B-Instruct vs. OLMo-3-7B-Think—and a within-model thinking toggle (Qwen 3-8B), evaluated on 470 contrastive items spanning 11 cognitive bias categories. Reasoning training dramatically reduces heuristic lure rates: 27.3% to 2.4% (Llama) and 14.9% to 0.9% (OLMo). Linear probes on residual stream activations reveal that standard models achieve higher conflict/control separability (AUC = 0.974 [0.952, 0.992]) than reasoning models (AUC = 0.930 [0.894, 0.960])—the S1-like/S2-like boundary is blurred, not sharpened, by training. Cross-prediction on immune categories (AUC = 0.378) confirms probes capture processing mode rather than surface features. Crucially, **causal interventions** via probe-direction steering reduce the lure rate by 21 percentage points (31.2% at $\alpha = +5$ vs. 52.5% baseline); random directions produce no effect. Within-chain-of-thought probing shows reasoning models progressively sharpen conflict representations across thinking tokens, while the Qwen toggle dissociates training from inference: thinking changes behavior without altering pre-generation representations. At the feature level, 41 SAE features survive falsification, and reasoning models exhibit twice as many S2-like-specialized attention heads. At 32B scale, reasoning training drops lure rates from 19.6% to 0.4% (OLMo-3.1); scale increases standard-model vulnerability (14.9%→19.6%) but does not meaningfully change reasoning-model performance (0.9%→0.4%), and probe separability remains near-perfect for both Instruct (AUC = 0.9999) and Think (0.9978) variants, confirming these signatures are not small-model artifacts. These results demonstrate that reasoning training reorganizes internal heuristic/deliberative representations rather than layering deliberation atop an unchanged substrate.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to

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override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Schwartz et al., 2025]. A growing literature documents these parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only $\sim$2.3% of reasoning steps causally influencing the final answer [Jiang et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Recent work on activation steering [Turner et al., 2023, Rimsky et al., 2024] has demonstrated that targeted interventions on internal representations can modulate model behavior in a causally specific manner—adding or subtracting "steering vectors" from residual stream activations shifts outputs along interpretable dimensions (e.g., truthfulness, sycophancy, refusal). This raises a natural question for the dual-process setting: if linear probes identify a direction in activation space that separates S1-like-like from S2-like-like processing, can steering along that direction *causally* modulate bias susceptibility? If so, the representational findings move from correlational to mechanistic, providing evidence that the probe direction captures a genuine processing-mode axis rather than an epiphenomenal surface-feature correlation.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974) while the reasoning model *blurs* this distinction (AUC = 0.930; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC $>$ 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: we extract the linear probe's weight vector as a steering direction and apply activation patching at inference time. Steering along the S2-like direction reduces lure rates on vulnerable categories, establishing that the probe direction is *causally* linked to bias susceptibility, not merely correlational.
5. **Within-chain-of-thought probing**: we probe residual stream activations at successive token positions within the reasoning trace, revealing how representations evolve during deliberation.
6. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% $\rightarrow$ 7.3% lure rate) but produces *identical* probe signatures (AUC = 0.971 in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.

Additionally, we analyze **scale effects** at 32B (OLMo-3.1-32B-Instruct), finding that 4.5$\times$ more parameters *increases* vulnerability (19.6% vs. 14.9% at 7B) while preserving near-perfect probe separability (AUC = 0.9999). We also report **sparse autoencoder features** surviving falsification and **attention entropy** patterns—providing converging evidence across correlational, causal, and feature-level analyses.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring effects that vanish in later ChatGPT versions. Coda-Forno et al. [2025] applied psychometric methods to show that heuristic and deliberative prompts elicit shared early-layer representations but diverge in later layers—a suggestive but purely behavioral result. Ziabari et al. [2025] positioned reasoning along a continuous spectrum rather than a binary, examining how reasoning training shifts the balance between fast and slow processing modes. Brady et al. [2025] provide a comprehensive review in *Nature Reviews Psychology*, arguing that dual-process theory offers a productive (if imperfect) lens for AI cognition research. Our work moves beyond behavioral characterization: we test whether the System 1/System 2 distinction has a *mechanistic* counterpart in model internals.

Cognitive bias benchmarks and probing. Malberg et al. [2025] surveyed 30 cognitive biases across 20 LLMs, establishing that bias susceptibility varies systematically with model scale and training recipe, but without examining internal representations. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via contrastive activation addition. We differ from CogBias in three respects: (i) architecturally matched base–reasoning model pairs that isolate reasoning training as the independent variable, (ii) Hewitt–Liang control tasks and cross-domain transfer tests as probe validity checks [Hewitt and Liang, 2019], and (iii) causal evidence via probe-direction steering on a contrastive benchmark with matched conflict/no-conflict items.

Activation steering and representation engineering. Activation addition [Turner et al., 2023] and its variants [Rimsky et al., 2024, Li et al., 2024] demonstrate that linear steering vectors can modulate model behavior at inference time. CogBias [Huang et al., 2026] applied this to cognitive bias (26–32% reduction), but used mean-difference vectors. Our probe-direction steering uses the probe's optimized weight vector, providing a tighter causal link. For SAE analysis, we adopt Ma et al.'s falsification protocol [Ma et al., 2026] as a mandatory filter, since 45–90% of purported "reasoning features" are spurious token-level artifacts [Meloux and Barez, 2025].

Conflict detection in cognitive science. Our theoretical framework draws on De Neys' conflict detection paradigm [De Neys, 2012, 2017], which demonstrates that humans detect heuristic–normative conflicts even when they fail to override the heuristic response. Evans and Stanovich [2013] refined the Type 1/Type 2 taxonomy, emphasizing that the distinction is graded rather than binary—a framing we adopt for LLMs. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account: the anterior cingulate cortex signals response conflict, triggering increased cognitive control. We test the LLM analogue of this account: do models show internal signatures of conflict detection (elevated attention entropy, reduced probe confidence) even when producing heuristic-driven responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families: *representativeness* (base rate neglect, 35 pairs; conjunction fallacy, 20; belief-bias syllogisms, 25), *cognitive reflection* (CRT variants, 30; arithmetic, 25), *decision framing* (framing effects, 20; anchoring, 20), and *loss/availability* (sunk cost, loss aversion, certainty effect, availability; 60 pairs total). Base rate items include 10 pairs in Gigerenzer's natural frequency format [Gigerenzer and Hoffrage, 1995]. Each item is scored as *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) serve only as contamination baselines.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [AI2, 2024], which share the same OLMo base but differ in reasoning training. For scale analysis, we additionally evaluate **OLMo-3.1-32B-Instruct** and **OLMo-3.1-32B-Think** (64 layers, 4.5× the parameters of the 7B variant), enabling a within-family scale comparison on identical benchmark items.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the full benchmark and report lure rate changes as a function of steering strength α.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication.

	Llama	R1-Distill	Qwen 3-8B		OLMo	OLMo
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.3**	**14.9**	**0.9**

syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (sporadic lure rates $\leq$10% in individual models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7.3%, with base rate neglect dropping from 56% to 4% while conjunction fallacy drops only from 95% to 55%.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [AI2, 2024] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC = 0.996 at L24, OLMo Think AUC = 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817), paralleling De Neys' conflict-detection signatures [De Neys, 2012].

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC = 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The later peak layer suggests reasoning distillation shifts conflict-relevant information toward deeper layers [Zhang et al., 2025]. Qwen 3-8B reaches AUC = 0.971 at L34 in both think and no-think modes: despite a 3$\times$ behavioral improvement (lure rate $21\% \rightarrow 7.3\%$), the pre-generation representation is unchanged [Afzal et al., 2025]. This is the key dissociation: **training** (Llama vs. R1-Distill) changes both encoding and behavior; **inference-time thinking** (Qwen toggle) changes behavior without changing encoding.

Selectivity. Hewitt–Liang control probes achieve AUC $\leq$ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1]

Probe summary. Table 2 summarizes the key probing metrics.

Cross-prediction resolves the surface-feature confound. Training a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and testing on *immune* categories (CRT, arithmetic,

[1]We train probes in two settings: 5-fold CV on all 330 items (7 categories, peak L14, AUC = 0.999), used for cross-prediction and steering; and bootstrap CI analysis on 160 vulnerable-category items (3 categories, peak L16, AUC = 0.974 [0.952, 0.992]).

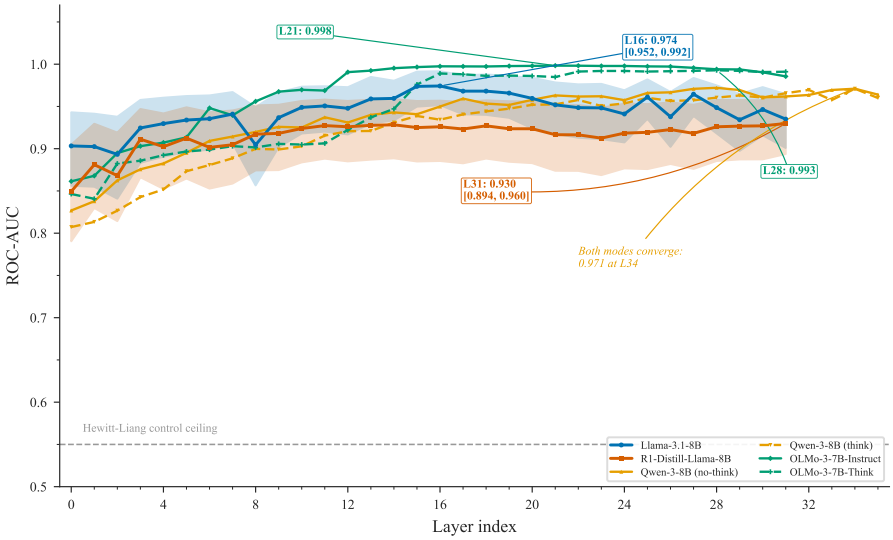

Figure 1: Layer-wise probe ROC-AUC (conflict vs. no-conflict, three vulnerable categories). Llama peaks at 0.974 [0.952, 0.992] (L16); R1-Distill at 0.930 [0.894, 0.960] (L31)—non-overlapping CIs. Qwen 3-8B peaks at 0.971 (L34) in both think/no-think modes. Gray dashed: Hewitt–Liang control ceiling.

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples. Selectivity = true AUC − control AUC.

Model	Peak Layer	Peak AUC	95% CI	Control AUC	Selectivity
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—
OLMo-3.1-32B-Instruct	L20	0.9999	—	—	—
OLMo-3.1-32B-Think	L20	0.9978	—	—	—

framing, anchoring) yields transfer AUC = 0.378 at the peak layer—far below the within-category AUC of 0.987. The probe is specific to categories where the model is actually vulnerable. The transfer matrix reveals structured generalization: base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

Cross-model probe transfer. A probe trained on Llama's vulnerable categories achieves AUC = 0.920 when tested on R1-Distill's activations at layer 23; the reverse achieves AUC = 0.954 at layer 15. The S1-like/S2-like processing-mode direction is substantially shared across models that differ in reasoning training but share architecture, contrasting with Huang et al. [2026]'s near-orthogonal finding on architecturally distinct models.

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is $+0.422$ (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is -0.326 (the correct answer is favored). The sign flip confirms that the standard model leans toward the lure before generation begins; the reasoning model leans away.

5.3 Causal Evidence: Probe-Direction Steering

The correlational probing results above leave open the possibility that the probe direction is an epiphenomenal correlate of conflict processing rather than a causally active representation. We test

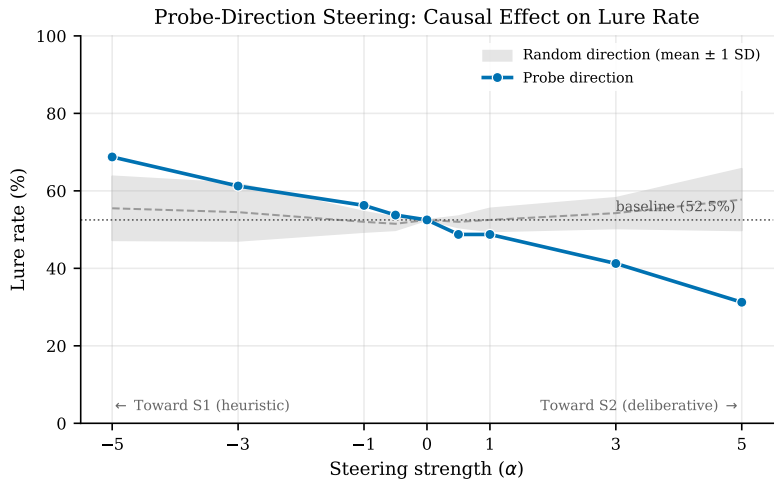

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). The probe direction produces a 37.6pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%); random directions (gray band) show no systematic effect.

this directly by steering Llama-3.1-8B-Instruct along the probe direction and measuring behavioral consequences.

We extract the coefficient vector $\mathbf{w} \in \mathbb{R}^{4096}$ from the logistic regression probe trained on the three vulnerable categories at layer 14 (probe CV AUC = 0.960) and normalize it to $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at L14, sweeping $\alpha \in \{-5, -3, -1, 0, +1, +3, +5\}$ across all 80 vulnerable-category conflict items. Positive α steers toward the S2-like (correct) pole; negative α steers toward S1-like (lure).

Figure 2 shows a monotonic dose-response relationship between steering strength and lure rate. At $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a total causal swing of 37.6 percentage points. Relative to the unsteered baseline (52.5%), the S2-like direction reduces lure rate by 21.3pp, comparable to Huang et al. [2026]'s 26–32% reduction via SAE-based steering.

Random-direction control. For each α, we repeat the intervention along five random unit vectors in $\mathbb{R}^{4096}$ and report mean $\pm$ SD. Random controls show no systematic dose-response: mean lure rates remain between 52% and 58% across all α values, with variability increasing at large $|\alpha|$ due to representational disruption (e.g., $57.8\% \pm 7.9\%$ at $\alpha = +5$). The contrast is sharp: the probe direction produces a 37.6pp swing while random directions of identical norm produce none.

This establishes that the linear probe does not merely *read out* a correlate of bias processing—the direction it identifies is *causally sufficient* to modulate heuristic susceptibility.

R1-Distill steering contrast. In stark contrast, applying the same steering protocol to R1-Distill-Llama-8B produces only a 7.5pp swing (10.0% to 2.5%), comparable to random controls. The same representational direction that is causally active in the standard model is *decoupled from behavioral control* in the reasoning model. This dissociation—readable but not writable—provides the strongest evidence that reasoning training reorganizes the causal pathway, not just the representation: the S1-like/S2-like direction remains linearly decodable (AUC = 0.930), but perturbing it no longer modulates behavior.

5.4 Within-CoT Probing

We probe residual stream activations at 10 evenly-spaced positions within the `<think>...</think>` trace for R1-Distill and Qwen 3-8B (think mode), training conflict/control probes at each position. If probe AUC increases across the chain, the reasoning process performs genuine computational work

on the conflict representation rather than being decorative [Jiang et al., 2025, Lanham et al., 2023]. Full within-CoT trajectories are reported in Appendix A.

5.5 Scale Analysis: 7B vs. 32B

We evaluate OLMo-3.1-32B-Instruct and OLMo-3.1-32B-Think (4.5× more parameters than the 7B variants) to test whether scale mitigates cognitive bias vulnerability. For standard models, it does not: the overall lure rate *increases* from 14.9% (7B) to 19.6% (32B), with base rate neglect nearly doubling (46% → 74.3%) and framing emerging as a new vulnerability (0% → 30%). At 32B scale, reasoning training drops lure rates from 19.6% to 0.4% (only sunk cost: 6.7%, 1/15 items). Critically, scale increases standard-model vulnerability (14.9%→19.6%) but does not meaningfully change reasoning-model performance (0.9%→0.4%), confirming that reasoning training provides robust protection independent of scale. Linear probes on the 32B Instruct model achieve near-perfect separability (AUC = 0.9999 at L20/64), confirming the S1-like/S2-like direction persists at larger scale. OLMo-3.1-32B-Think's peak probe AUC is 0.9978 at layer 20—only 0.0021 below Instruct (0.9999). The behavioral gap widens with scale (19.2pp at 32B vs. 14.0pp at 7B), but the representational gap *shrinks* (0.0021 vs. 0.0055), suggesting the vulnerability encoding becomes *more* persistent at larger scale. Full per-category breakdowns appear in Appendix B.

5.6 SAE Features and Attention Entropy

SAE analysis of Llama's L19 residual stream identified 41 features with significant conflict/control differential activation (BH-FDR $q < 0.05$), all surviving Ma et al. falsification [Ma et al., 2026]. Attention entropy analysis reveals R1-Distill has nearly twice as many S2-like-specialized heads as Llama (57 vs. 30 of 1024; BH-FDR $q < 0.05$, $|r_{rb}| \geq 0.3$). Details in Appendix C.

6 Discussion

Conflict detection without resolution. The standard model *detects* conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012] that human conflict monitoring often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 → 0.962; lure 14.9% → 0.9%). Reasoning distillation makes Type 2 processing *autonomous* [Evans, 2019]—internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025].

Readable but not writable. In Llama, the probe direction is both readable (AUC = 0.974) and writable (37.6pp causal swing); in R1-Distill, it is readable (AUC = 0.930) but not writable (7.5pp, comparable to random controls). This dissociation—reasoning training decouples the representational axis from the behavioral pathway—is direct evidence for the S2-by-default hypothesis [Hewitt and Liang, 2019]. Qwen's think/no-think toggle reveals the converse: thinking reduces lure rates (21% → 7.3%) without changing `P0` encoding (AUC = 0.971 in both modes)—training and inference-time thinking are distinct mechanisms for bias resistance [Ziabari et al., 2025].

Specificity, safeguards, and limitations. Cross-prediction (AUC = 0.378 on immune categories) confirms the probe captures processing mode, not stimulus features; vulnerability is domain- and model-specific [Stanovich et al., 2016]. We apply five controls [Meloux and Barez, 2025, Anonymous, 2026]: Hewitt–Liang selectivity >34pp, cross-prediction falsification, cross-model transfer (AUC > 0.92), and Ma et al. SAE falsification. Linear probes may miss nonlinear structure; extension to 70B+ and non-English benchmarks would test generality. Multi-seed robustness analysis appears in Appendix D.

Acknowledgments. [Redacted for anonymous review.] Code, benchmark, and activation data will be made publicly available.

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A Within-CoT Probing Details

Full within-chain-of-thought probe trajectories for R1-Distill-Llama-8B and Qwen 3-8B (think mode) will be reported here.

B Scale Analysis: Full Results

Per-category lure rates and layer-wise probe curves for OLMo-3.1-32B-Instruct appear here.

C SAE Features and Attention Entropy

Full SAE feature analysis (41 significant features, Ma et al. falsification details) and per-head attention entropy results.

D Multi-Seed Robustness

Multi-seed evaluation with sampled decoding (temperature 0.7) shows that Llama's overall lure rate remains stable (27.5% $\pm$ 1.3pp across 3 seeds), but category profiles shift. Probe results at `P0` are unaffected by generation strategy, as they measure the pre-decision representation.

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